



Normalization in Style Transfer

LAB Team

Style Transfer



Content? Style?



- ***Similar in content:***

=> high-level features as extracted by a trained classifier are close in Euclidian distance.

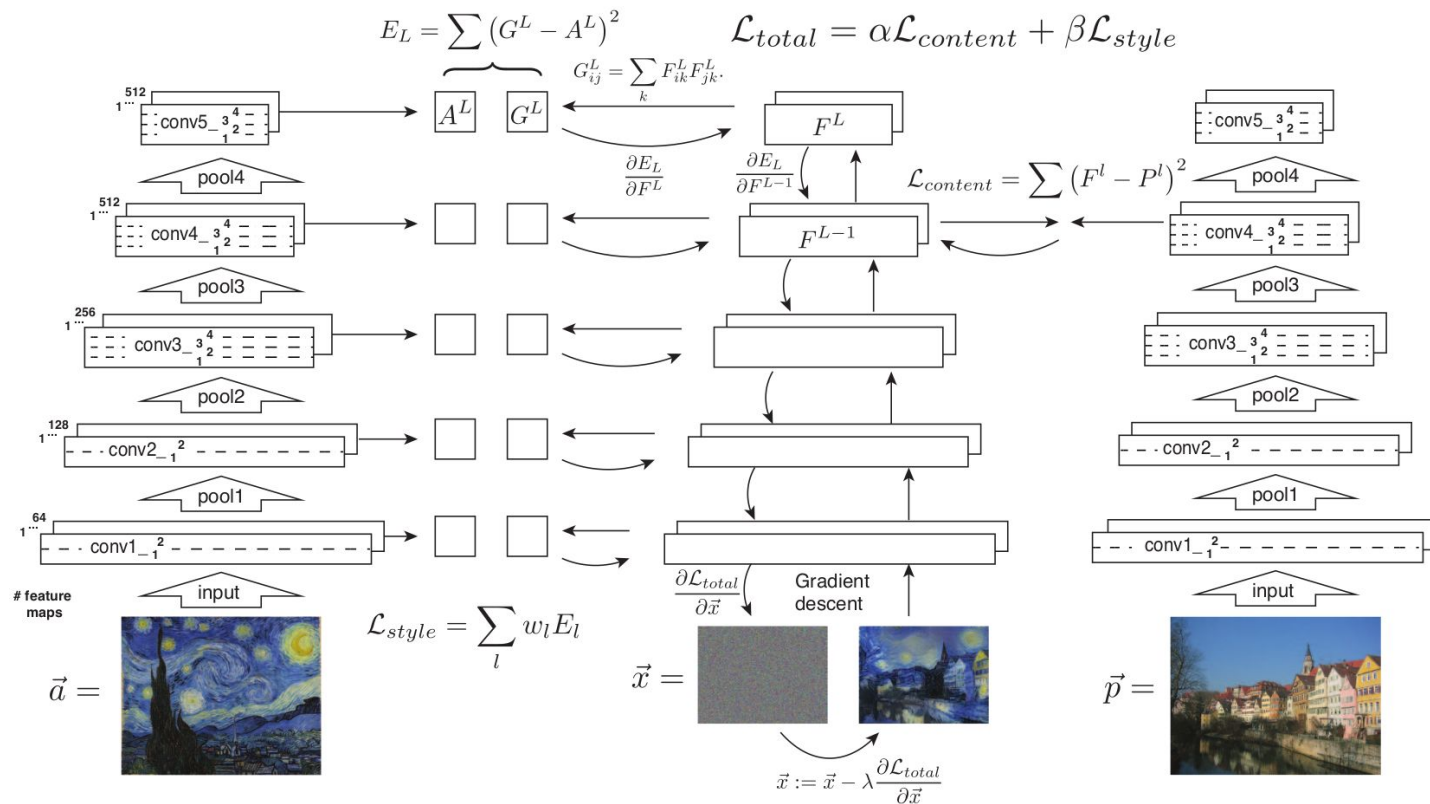
- ***Similar in style:***

=> low-level features as extracted by a trained classifier share the same statistics

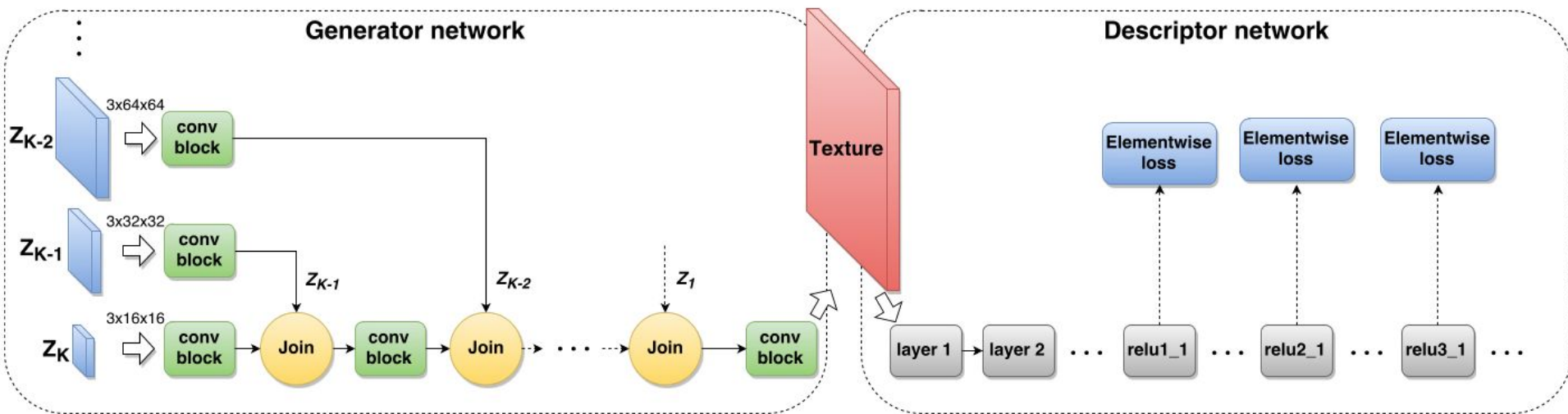
or

=> the difference between the features' Gram matrices has a small Frobenius norm.

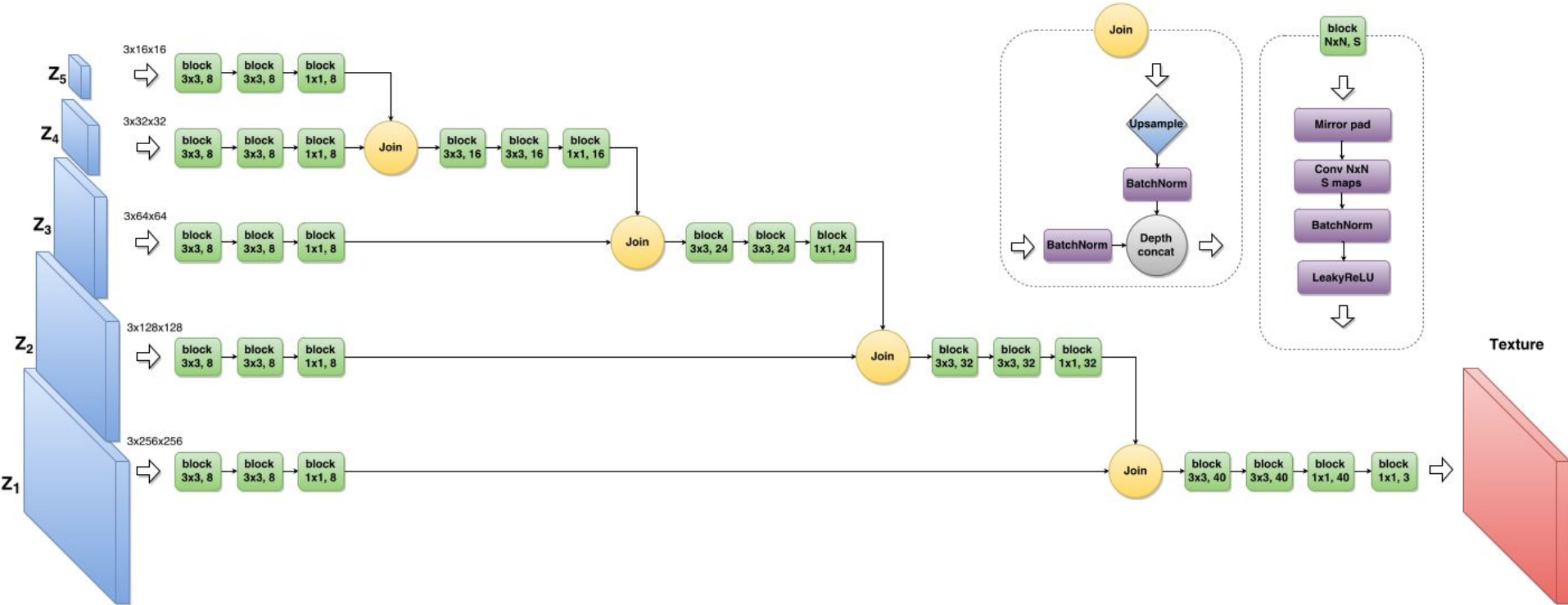
Style Transfer



Texture Network



Texture Network



Batch Norm



$$\text{BN}(x) = \gamma \left(\frac{x - \mu(x)}{\sigma(x)} \right) + \beta$$

$$\mu_c(x) = \frac{1}{NHW} \sum_{n=1}^N \sum_{h=1}^H \sum_{w=1}^W x_{nchw}$$

$$\sigma_c(x) = \sqrt{\frac{1}{NHW} \sum_{n=1}^N \sum_{h=1}^H \sum_{w=1}^W (x_{nchw} - \mu_c(x))^2 + \epsilon}$$

Batch Norm



- Uses mini-batch statistics during training
 - Uses popular statistics during inference
- => Discrepancy between training and inference
- => ***Batch Re-Norm or other Norms***

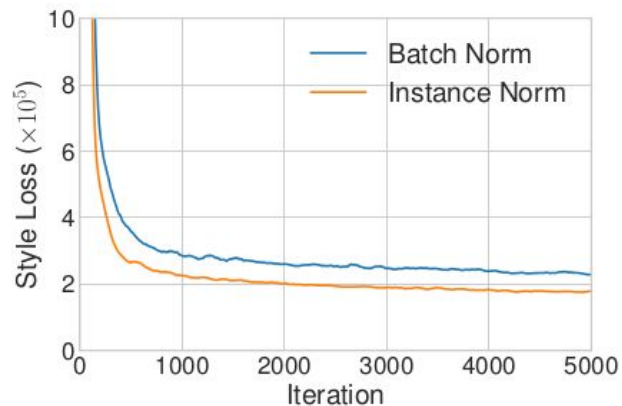
Instance Norm

$$\text{IN}(x) = \gamma \left(\frac{x - \mu(x)}{\sigma(x)} \right) + \beta$$

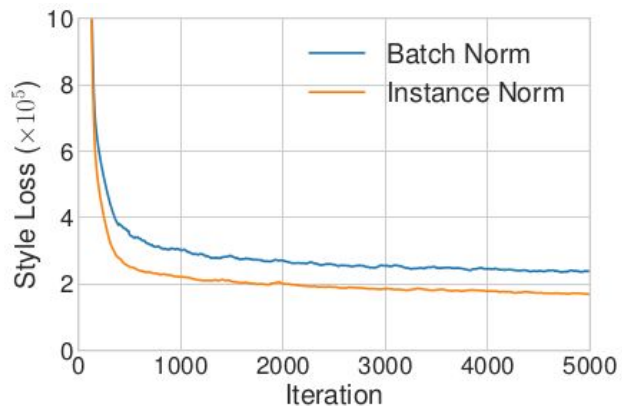
$$\mu_{nc}(x) = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W x_{nchw}$$

$$\sigma_{nc}(x) = \sqrt{\frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W (x_{nchw} - \mu_{nc}(x))^2 + \epsilon}$$

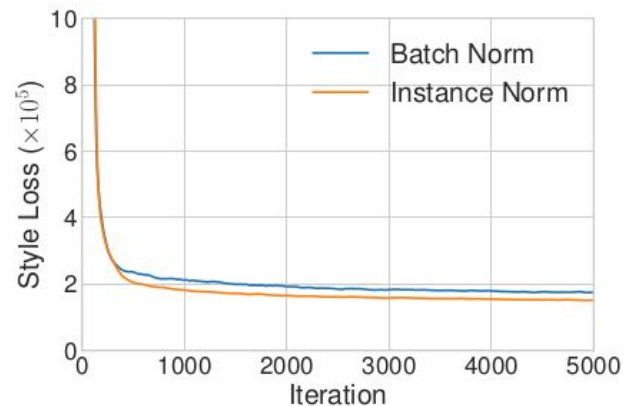
BN vs IN



(a) Trained with original images.



(b) Trained with contrast normalized images.



(c) Trained with style normalized images.

BN vs IN



Content

StyleNet IN (ours)

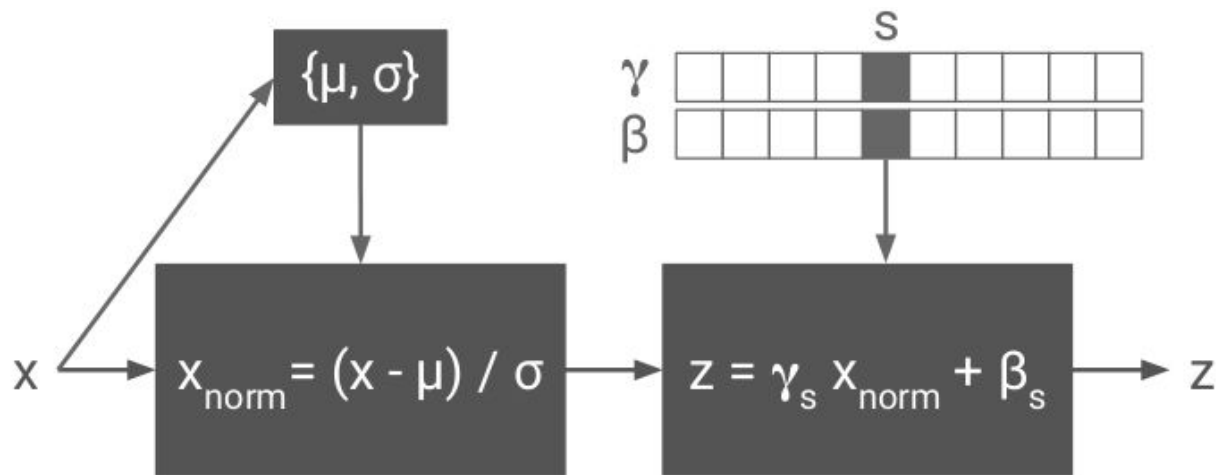
StyleNet BN

Gatys et al.

Style

Conditional Instance Norm

$$\text{CIN}(x; s) = \gamma^s \left(\frac{x - \mu(x)}{\sigma(x)} \right) + \beta^s$$

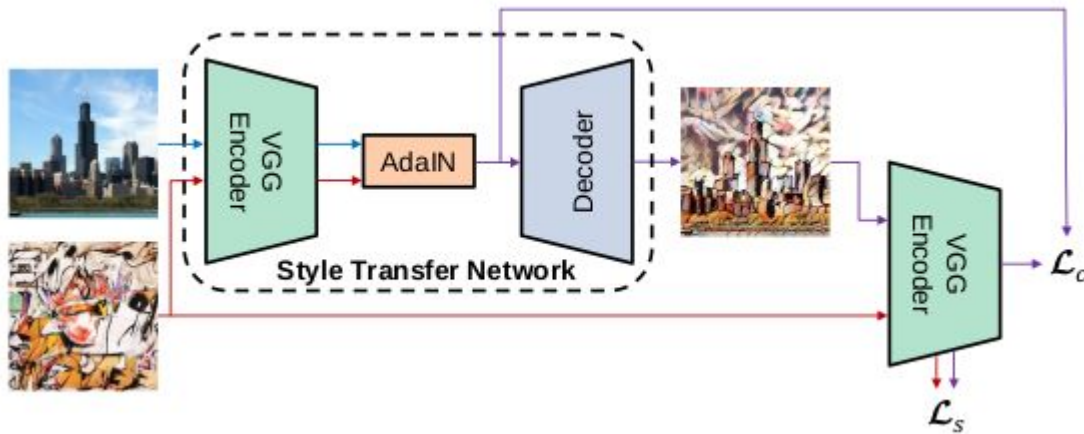


Conditional Instance Norm



Adaptive Instance Norm

$$\text{AdaIN}(x, y) = \sigma(y) \left(\frac{x - \mu(x)}{\sigma(x)} \right) + \mu(y)$$

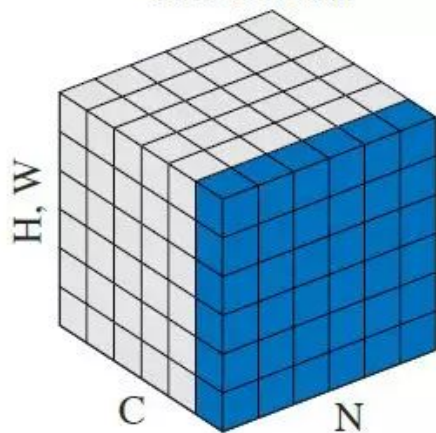


Layer Norm

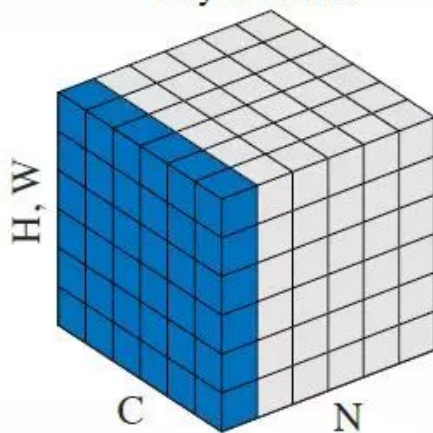
$$\mu^l = \frac{1}{H} \sum_{i=1}^H a_i^l \quad \sigma^l = \sqrt{\frac{1}{H} \sum_{i=1}^H (a_i^l - \mu^l)^2}$$

Layer Norm

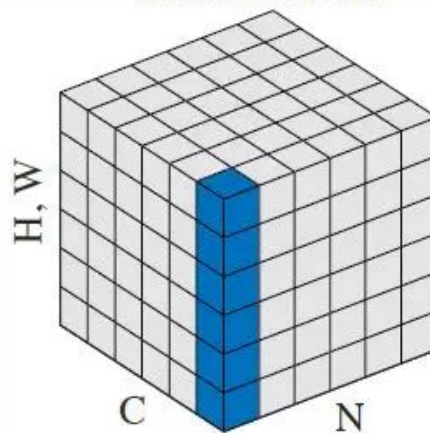
Batch Norm



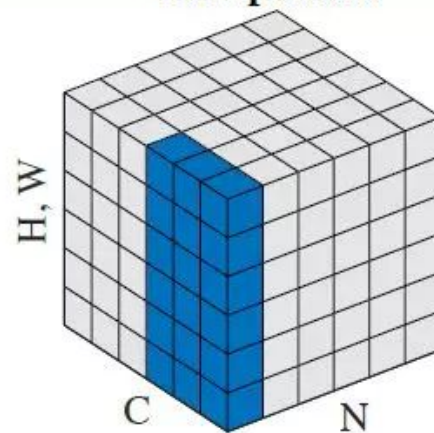
Layer Norm



Instance Norm



Group Norm



Batch Instance Norm

$$\mathbf{y} = \left(\rho \cdot \hat{\mathbf{x}}^{(B)} + (1 - \rho) \cdot \hat{\mathbf{x}}^{(I)} \right) \cdot \gamma + \beta, \quad \rho \in [0, 1]^C$$

Additional learnable parameters ρ

$$\frac{\partial \ell}{\partial \rho_c} = \gamma_c \sum_N \sum_H \sum_W \left(\hat{x}_{nchw}^{(B)} - \hat{x}_{nchw}^{(I)} \right) \frac{\partial \ell}{\partial y_{nchw}},$$

$$\rho \leftarrow \text{clip}_{[0,1]} (\rho - \eta \Delta \rho)$$

Batch Instance Norm

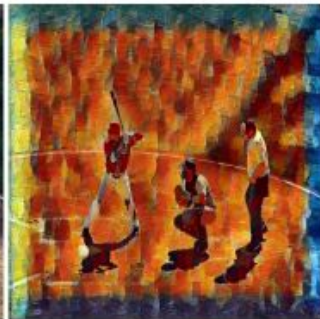
Style



Content



BN



IN



BN+IN



BIN



Adaptive Layer Instance Norm

$$AdaLIN(a, \gamma, \beta) = \gamma \cdot (\rho \cdot \hat{a}_I + (1 - \rho) \cdot \hat{a}_L) + \beta$$

$$\hat{a}_I = \frac{a - \mu_I}{\sqrt{\sigma_I^2 + \epsilon}}, \hat{a}_L = \frac{a - \mu_L}{\sqrt{\sigma_L^2 + \epsilon}},$$

$$\rho \leftarrow clip_{[0,1]}(\rho - \tau \Delta \rho)$$